One-Way Between-Subjects ANOVA Demo

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One-way Between-Subjects ANOVA | Research Context Prompt

A developmental psychologist is looking for a Grad student who can help them with their research project looking into adolescent self-esteem and its relationship with parenting style. There were three parenting styles: (1) `authoritative` (high expectations with support), (2) `authoritarian` (high expectations with little support), and (3) `permissive` (low expectations with support).

Data on 90 teenagers self-esteem levels and the parenting style they experienced (30 per style) has been collected. The research question of interest is: Does parenting style differences impact teenage self-esteem?

One-way Between-Subjects ANOVA | Data

```
library(psych)
library(tidyverse)
library(ggrain)
library(jmv)
library(ggpubr)
library(apaTables)
library(rstatix)

dat_A <- read.csv("One-way ANOVA.A.csv")
dim(dat_A)
```

```
## [1] 90  3
```

```
head(dat_A)
```

```
##   Teenager Style Self_esteem
## 1        1     1       34.88
## 2        2     1       38.73
## 3        3     1       35.34
## 4        4     1       34.53
## 5        5     1       39.22
## 6        6     1       37.65
```

One-way Between-Subjects ANOVA | Descriptive Stats

```
# Descriptive Stats
describe(dat_A[,2:3])
```

```
##             vars  n  mean   sd median trimmed  mad   min  max range  skew
## Style          1 90  2.00 0.82   2.00    2.00 1.48  1.00  3.0  2.00  0.00
## Self_esteem    2 90 34.46 2.50  34.79   34.48 2.31 28.58 39.8 11.22 -0.11
##             kurtosis   se
## Style          -1.53 0.09
## Self_esteem    -0.37 0.26
```

One-way Between-Subjects ANOVA | Grouped Descriptive Stats

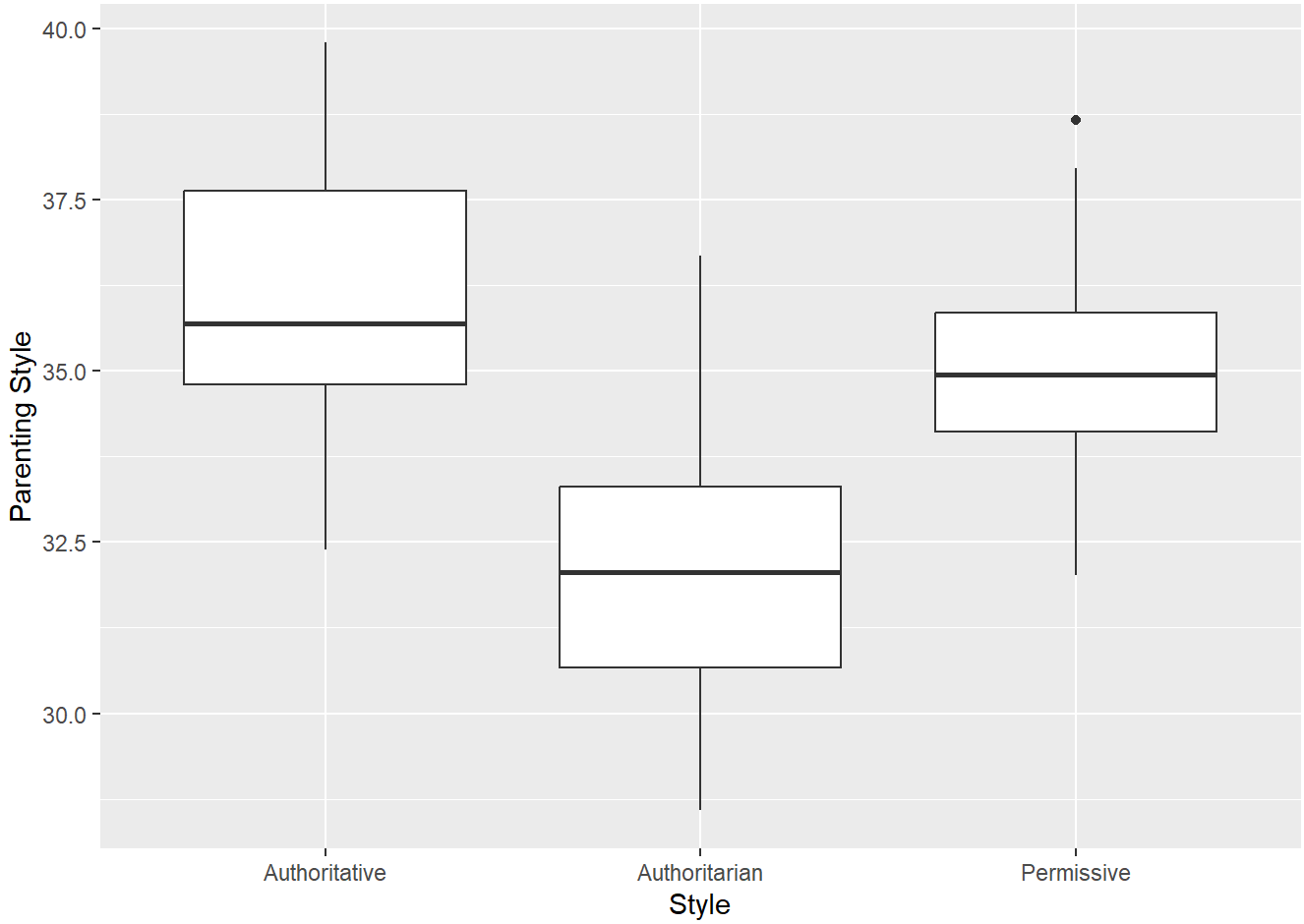
```
dat_A$Style<-factor(dat_A$Style,
                    labels = c("Authoritative","Authoritarian","Permissive"))

# Descriptive Stats
describeBy(dat_A$Self_esteem,
           dat_A$Style,
           mat=TRUE)
```

```
##     item        group1 vars  n     mean       sd median  trimmed      mad   min
## X11    1 Authoritative    1 30 36.26967 1.911749 35.690 36.20000 1.905141 32.39
## X12    2 Authoritarian    1 30 32.14300 1.986734 32.060 32.08917 2.142357 28.58
## X13    3    Permissive    1 30 34.96133 1.531965 34.945 34.91042 1.341753 32.02
##       max range      skew   kurtosis        se
## X11 39.80  7.41 0.2585508 -0.9602706 0.3490359
## X12 36.68  8.10 0.3020997 -0.6486497 0.3627263
## X13 38.67  6.65 0.2448705 -0.1540893 0.2796972
```

One-way Between-Subjects ANOVA | Univariate Visualizations

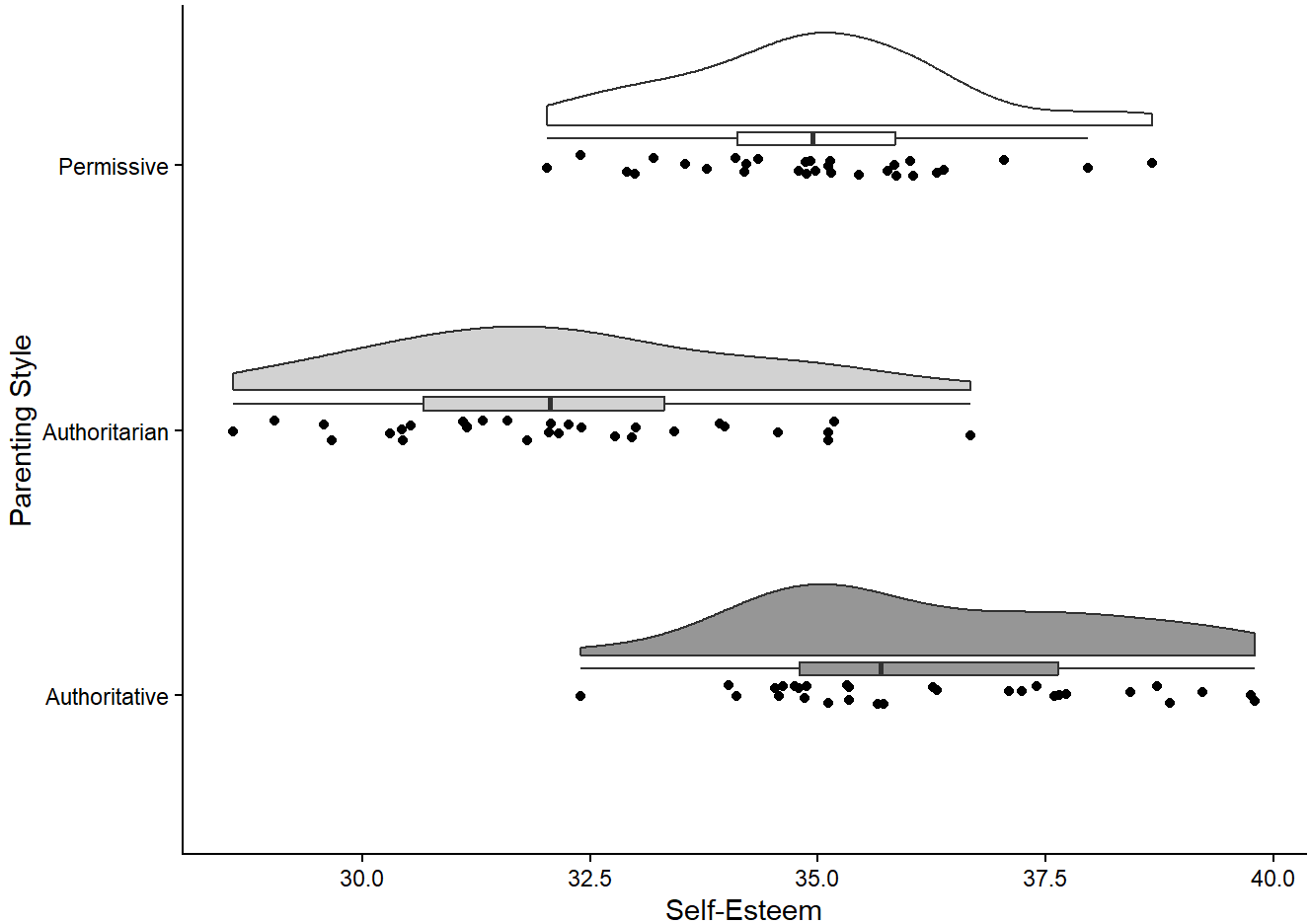
One-way Between-Subjects ANOVA | Grouped Boxplot

```
ggplot(data = dat_A,
       mapping = aes(y = Self_esteem, x =Style)) +
  geom_boxplot() +
    labs(y = "Parenting Style")
```

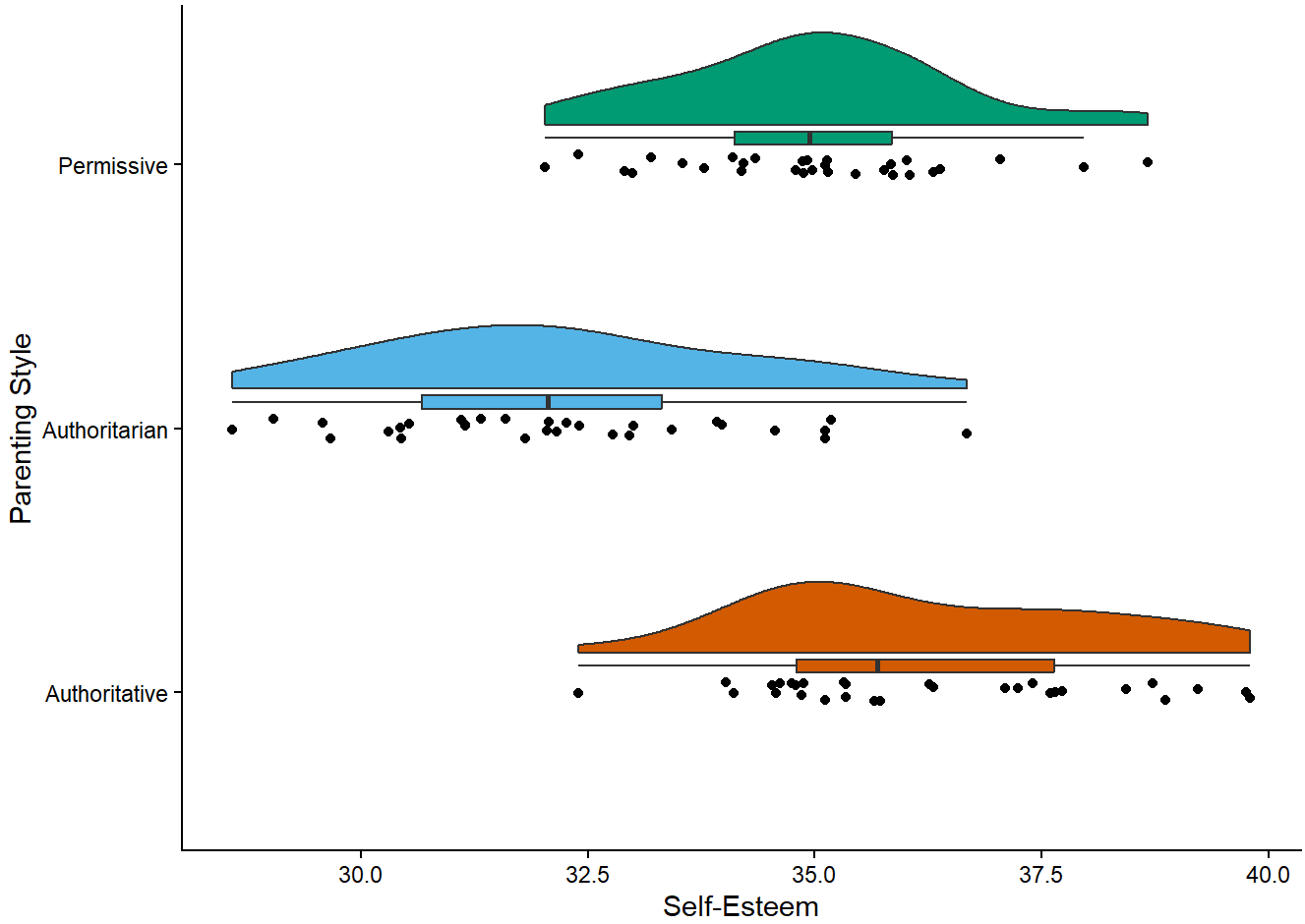


One-way Between-Subjects ANOVA | Raincloud Plot

```
ggplot(data = dat_A,
       aes(x = Style, y = Self_esteem,
           fill = Style))+
  geom_rain(show.legend=FALSE) +
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Self-Esteem")+
  xlab("Parenting Style")+
  theme_classic()
```




```
ggplot(data = dat_A,
       aes(x = Style, y = Self_esteem,
           fill = Style))+
  geom_rain(show.legend=FALSE) +
  coord_flip() +
  ylab("Self-Esteem")+
  xlab("Parenting Style")+
  scale_fill_manual(values=c("#D55e00","#56B4E9","#009E73"))+
  theme_classic()
```

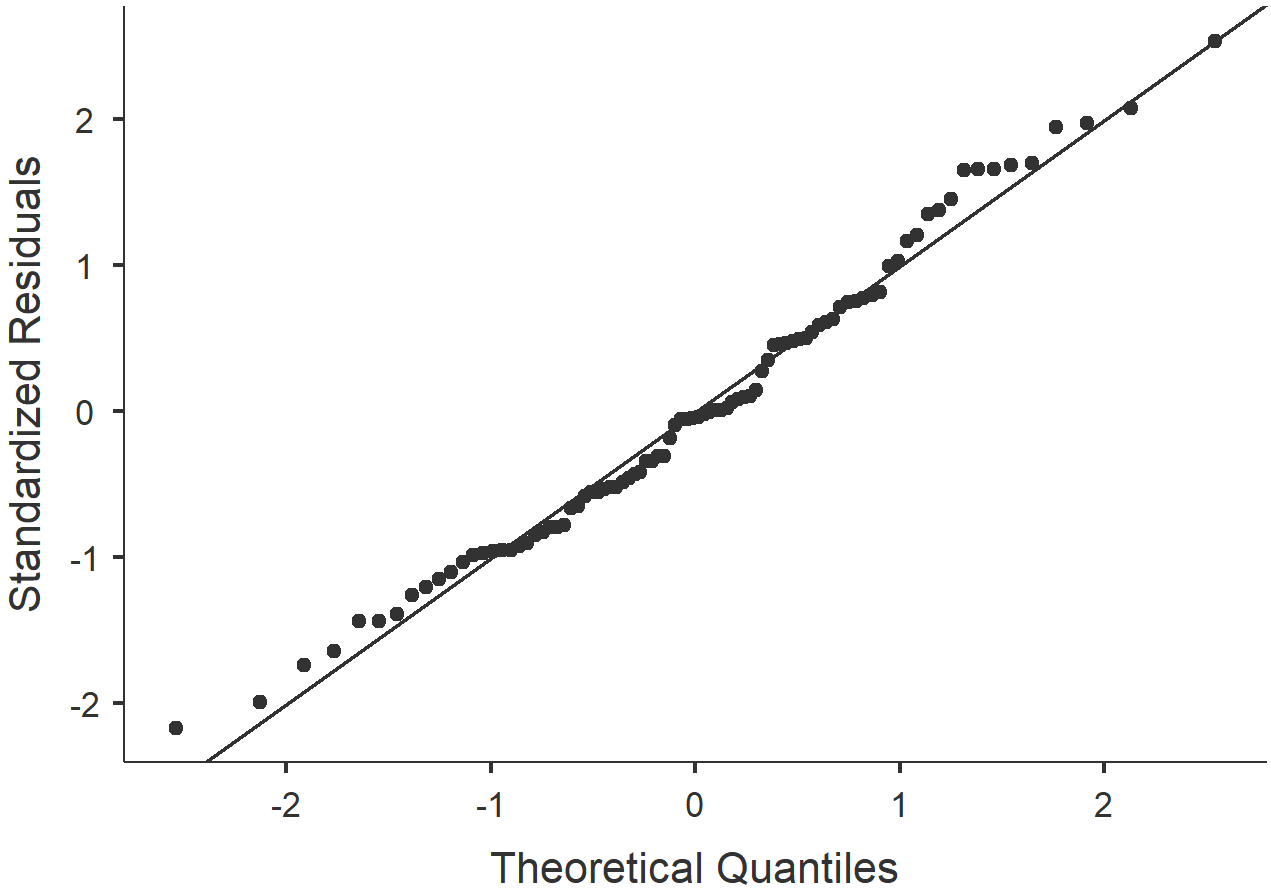


One-way Between-Subjects ANOVA | Omnibus Assumption Checks Code

```
# Omnibus Assumption Checking
ANOVA(data = dat_A,
      dep = 'Self_esteem',
      factors = c('Style'),
      homo = TRUE,
      norm = TRUE,
      qq = TRUE)
```

```
##
##  ANOVA
##
##  ANOVA - Self_esteem
##  ─────────────────────────────────────────────────────────────────────────────
##                  Sum of Squares    df    Mean Square    F           p
##  ─────────────────────────────────────────────────────────────────────────────
##    Style              266.8412     2     133.420583    40.23213    < .0000001
##    Residuals          288.5155    87       3.316270
##  ─────────────────────────────────────────────────────────────────────────────
##
##
##  ASSUMPTION CHECKS
##
##  Homogeneity of Variances Test (Levene's)
##  ─────────────────────────────────────────────
##    F           df1    df2    p
##  ─────────────────────────────────────────────
##    1.782551      2     87    0.1742966
##  ─────────────────────────────────────────────
##
##
##  Normality Test (Shapiro-Wilk)
##  ─────────────────────────────
##    Statistic    p
##  ─────────────────────────────
##    0.9846491    0.3708903
##  ─────────────────────────────
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



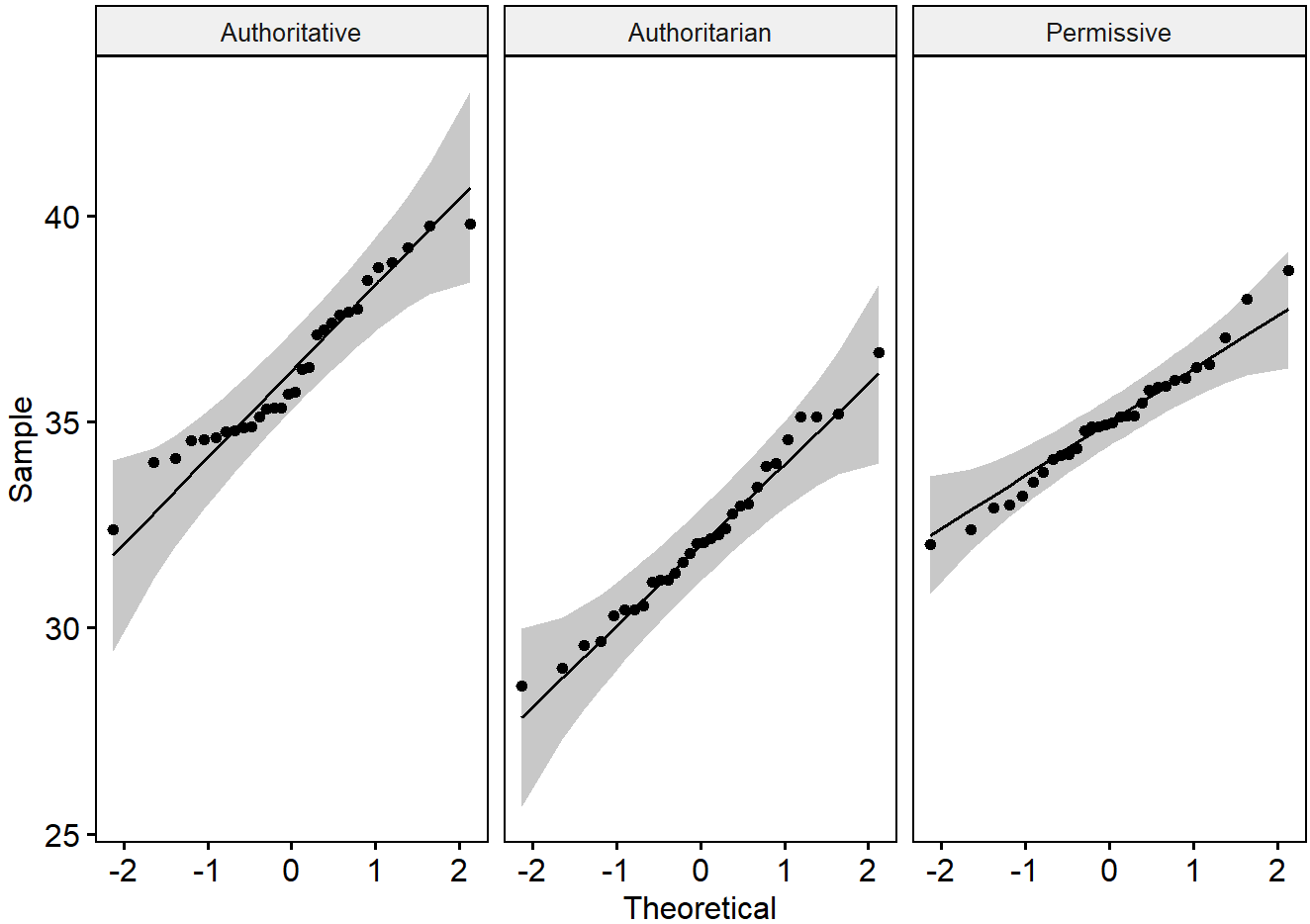
One-way Between-Subjects ANOVA | Group Level Assumption Checks

```
dat_A %>%
  group_by(Style) %>%
  shapiro_test(Self_esteem)
```

```
## # A tibble: 3 × 4
##   Style         variable    statistic     p
##   <fct>         <chr>           <dbl> <dbl>
## 1 Authoritative Self_esteem     0.949 0.157
## 2 Authoritarian Self_esteem     0.980 0.828
## 3 Permissive    Self_esteem     0.982 0.870
```

One-way Between-Subjects ANOVA | Group Level Assumption Checks

```
ggqqplot(dat_A,
         'Self_esteem',
         facet.by = "Style")
```



One-way Between-Subjects ANOVA | Rechecking Homogen. of Variance using `rstatix`

```
dat_A %>%
  levene_test(Self_esteem ~ as.factor(Style), center = mean)
```

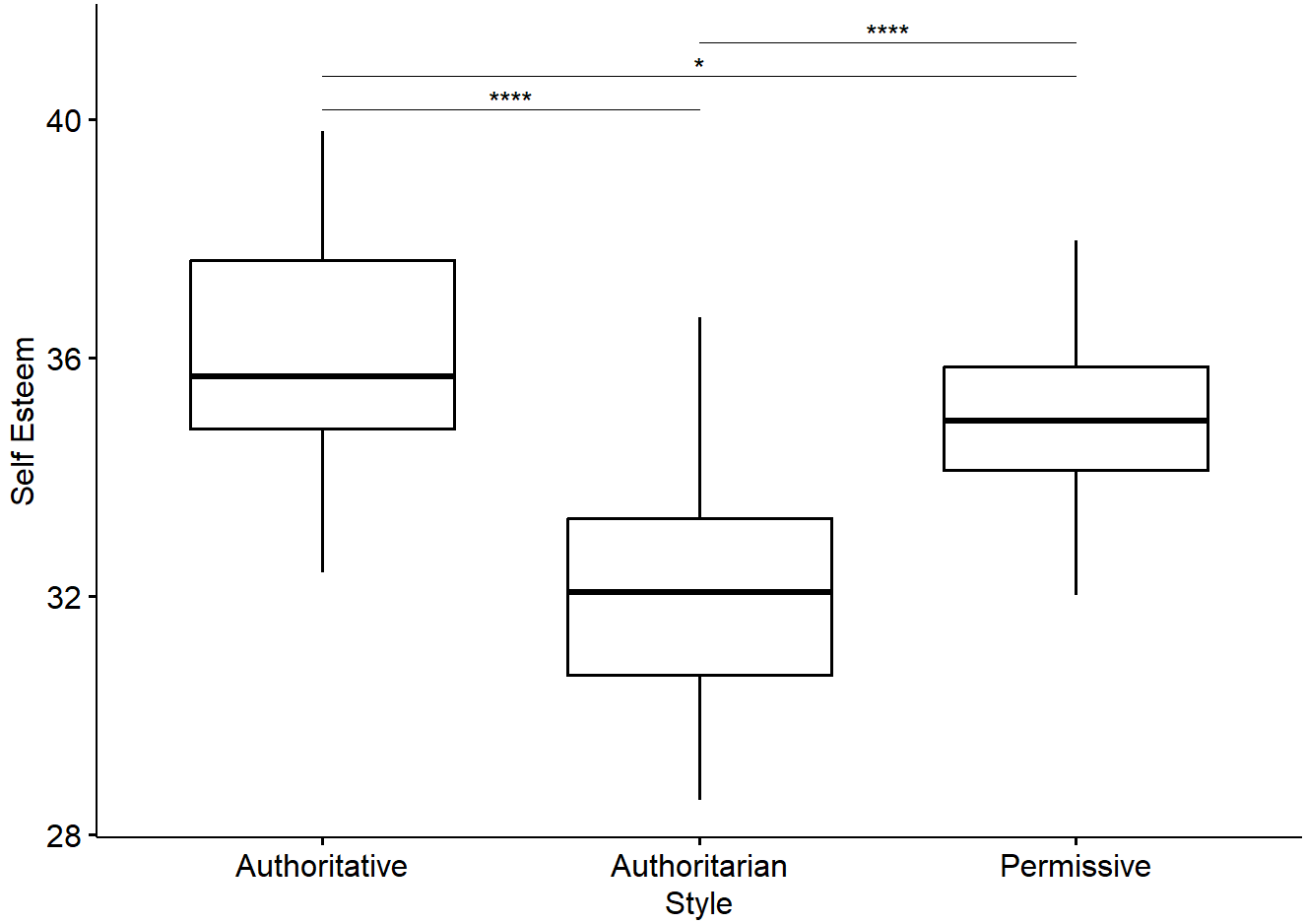
```
## # A tibble: 1 × 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     2    87      1.78 0.174
```

One-way Between-Subjects ANOVA | Conducting the ANOVA Code

```
options(digits = 3)
ANOVA(data = dat_A,
      dep = 'Self_esteem',
      factors = c('Style'),
      effectSize = 'eta',
      postHoc = 'Style',
      postHocCorr = 'bonf',
      postHocES = 'd',
      postHocEsCi = TRUE,
      emMeans = list(
       list('Style')),
      emmPlots = FALSE,
      emmTables = TRUE)
```

```
##
##  ANOVA
##
##  ANOVA - Self_esteem
##  ─────────────────────────────────────────────────────────────────────────────
##                  Sum of Squares    df    Mean Square    F       p         η²
##  ─────────────────────────────────────────────────────────────────────────────
##    Style                   267     2         133.42    40.2    < .001    0.480
##    Residuals               289    87           3.32
##  ─────────────────────────────────────────────────────────────────────────────
##
##
##  POST HOC TESTS
##
##  Post Hoc Comparisons - Style
##  ─────────────────────────────────────────────────────────────────────────────
```

```
# Pairwise Post-Hoc Tests (Bonferroni Correction)
pwc <- dat_A %>%
  pairwise_t_test(Self_esteem ~ Style, p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "Style")

# Visualization: box plots with p-values

ggboxplot(dat_A,
          x = "Style",
          y = "Self_esteem",
          #notch = TRUE,
          outlier.shape = NA) +
  stat_pvalue_manual(pwc,
                     hide.ns = TRUE,
                     tip.length = 0) +
  labs(y = "Self Esteem")
```



One-way Between-Subjects ANOVA | APA Style ANOVA Tables

```
lm<-lm(Self_esteem ~ Style, dat_A)
apa.aov.table(lm,
              table.number=1,
              conf.level=.95,
              type = 2)
```

```
##
##
## Table 1
##
## ANOVA results using Self_esteem as the dependent variable
##
##
##  Predictor     SS df     MS     F    p partial_eta2 CI_95_partial_eta2
##      Style 266.84  2 133.42 40.23 .000          .48         [.32, .59]
##      Error 288.52 87   3.32                                  [NA, NA]
##
## Note: Values in square brackets indicate the bounds of the 95% confidence interval for partia
l eta-squared
```

```
apa.1way.table(Style,
               Self_esteem,
               dat_A,
               table.number=2,
               show.conf.interval = TRUE)
```

```
##
##
## Table 2
##
## Descriptive statistics for Self_esteem as a function of Style.
##
##          Style     M       M_95%_CI   SD
##  Authoritative 36.27 [35.56, 36.98] 1.91
##  Authoritarian 32.14 [31.40, 32.88] 1.99
##     Permissive 34.96 [34.39, 35.53] 1.53
##
## Note. M and SD represent mean and standard deviation, respectively.
## LL and UL indicate the lower and upper limits of the 95% confidence interval
## for the mean, respectively.
## The confidence interval is a plausible range of population means that could
## have caused a sample mean (Cumming, 2014).
```